BS EN 50131 GRADE 2 & GRADE 3 INTRUDER ALARM SPECIFICATION

Jeweller 868MHz Encrypted Frequency Hopping vs 433MHz Jammer Vulnerabilities

Lead Engineer: Gary Pearce (25+ Years Experience • Direct: 07860 244 274)
Standards Compliance: BS EN 50131-1:2006+A2:2017 • PD 6662:2017 • Grade 2 & 3
Insurance Underwriting: £10,000,000 Public Liability Insurance • Dual SIM 4G + Gigabit LAN Signalling
Companion Field Debrief: tvandhomesecurity.blogspot.com/ajax-smart-alarms

1. The 433MHz Radio Jammer Vulnerability in UK Domestic Alarms

Commercial and residential burglaries across Yorkshire and Lancashire increasingly feature handheld £30 sub-GHz software-defined radios (Flipper Zero, CC1101 transmitters) tuned to the unencrypted 433.92MHz frequency. Legacy wireless alarm systems (Yale, ERA, uncertified DIY systems) operate on fixed 433MHz carrier waves without frequency hopping. An intruder broadcasting continuous RF noise blocks sensor transmissions completely; the central hub never receives the opening signal, and the outdoor siren fails to trigger.

Specification	Legacy DIY Wireless (Yale/ERA)	Ajax Jeweller Protocol (Grade 2)	Industrial Hardwired (Grade 3)
Operating Frequency	433.92 MHz (Fixed Carrier)	868.0 – 868.6 MHz (TDMA Multi-channel)	Hardwired (Solid Copper End-of-Line)
Encryption Standard	Unencrypted or Fixed 128-bit	AES-128 Floating Rolling Key	Encrypted Bus Line / Balanced Loop
Jamming Detection	None (Blind Jamming)	Instant Sub-Second Frequency Hop	Physical Cut / Short Circuit Alarm
Battery Reserve Life	1 to 2 Years (Alkaline)	Up to 7 Years (CR123A Lithium)	Mains 230V + 7Ah Lead Acid Backup
Photo Verification	None	MotionCam PhOD (0.15s Transmit)	IP Camera Integration RTSP

2. Dual-Path Signalling & MotionCam PhOD Evidence

Under PD 6662:2017 standards, a confirmed alarm requires verification across two distinct detection vectors to eliminate false dispatches. Ajax MotionCam PhOD detectors integrate dual-PIR optical triangulation with a CMOS snapshot camera that transmits visual evidence to Gary Pearce's monitoring station and the client's smartphone in less than 9 seconds over encrypted Wings radio protocol. Dual Micro-SIM cellular modules with roaming coverage across EE, Vodafone, and O2 ensure uninterrupted failover if broadband fiber lines are severed.

3. Engineer Deployment & Professional Specification

Every Ajax smart alarm system installed by ADI Communications includes stainless steel wall mounting, masonry rawlplugs, 90mph gale-tested outdoor StreetSirens (113dB sound pressure), and full compliance certificates for UK home insurance underwriters.

- **Full Technical Field Analysis:** tvandhomesecurity.blogspot.com/ajax-smart-alarms
- **Direct Engineer Contact:** Call Gary Pearce on **07860 244 274** for an on-site security survey.